Build guide 1: tools and steel

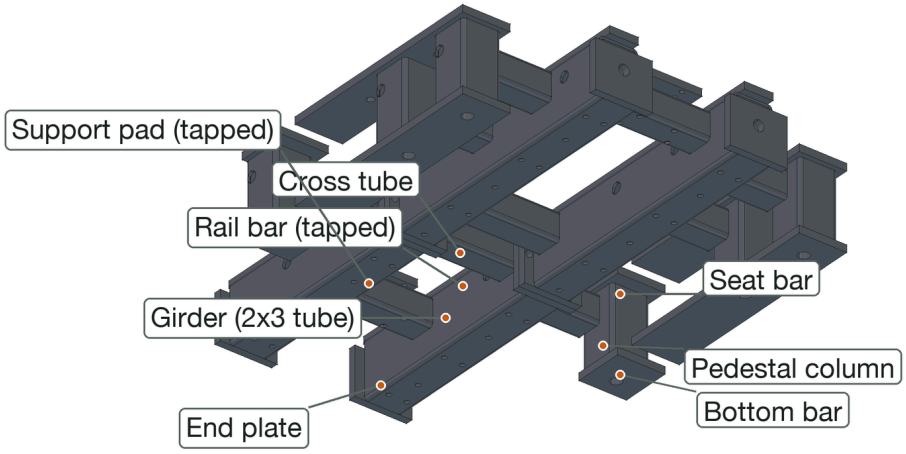
DIY CNC epoxy-granite mill · October 2026

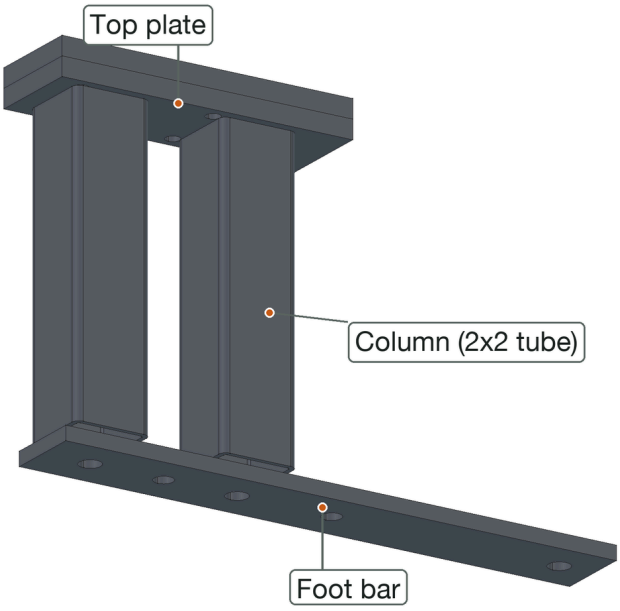
0. Tools, space and safety

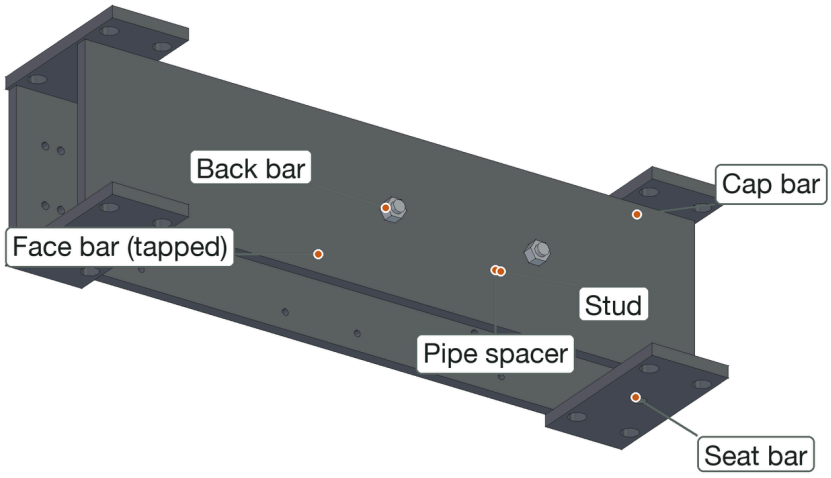
- **Space:** a level bench or floor area of about 1.2 x 1.5 m at 20-25 °C for a week. Epoxy cures slowly and badly below 18 °C.
- **Tools:** drill press, chop saw or hacksaw, angle grinder with flap disc, taps (M4, M5, M6), drill/mixer with a paddle (or a small cement mixer), kitchen scale + a 50 kg scale, vibration source (a table vibrator, a concrete poker, or a reciprocating sander with the pad removed pressed against the mold), straightedge (at least 600 mm, ideally ground), feeler gauges, dial indicator with a magnetic base, precision level, torque wrench.
- **Safety:** nitrile gloves and long sleeves for epoxy (skin sensitisation is permanent); an N95 dust mask when pouring the whiting; a P100 respirator when grinding or sanding cured EG (the quartz sand and gravel in it make silica dust; work wet if you can); eye protection when cutting steel. Mix in a ventilated space.
- **3D printing (PETG):** the mold kit (brackets, saddles, holders, bushings), the core posts, formers and fillers, the parts that get cast in (column rings, tube plugs, drain liner) and the machine's printed parts. All of it is in `print/`, split to fit a Bambu A1 mini, and listed in the print guide: 1.1 kg of PETG (sliced), inside two 1 kg rolls, and about 90 hours of printing. The base pocket cores are cut from one sheet of 2 in XPS foam board instead. Print one bracket and one saddle first and check them on your board; print a column ring early and test-fit it in your actual tube.
- **Foam cutting:** a fine-tooth saw or a hot-wire cutter, a long knife, a sanding block, packing tape and double-sided carpet tape.

1. Buy and prepare the steel

The frame's backbone is steel: a skeleton in each casting, bolted together from **stock flat bar and tube** and clamped with threaded rod, then cast into epoxy granite, which damps it and stiffens it further. Every joint between the castings, and everything that bolts on (rails, ballscrew supports), is steel to steel. Nothing is welded, nothing is custom-cut, and nothing is threaded into the epoxy granite.


Support pad (tapped)
Cross tube
Rail bar (tapped)
Seat bar
Girder (2x3 tube)
Pedestal column
Bottom bar
End plate

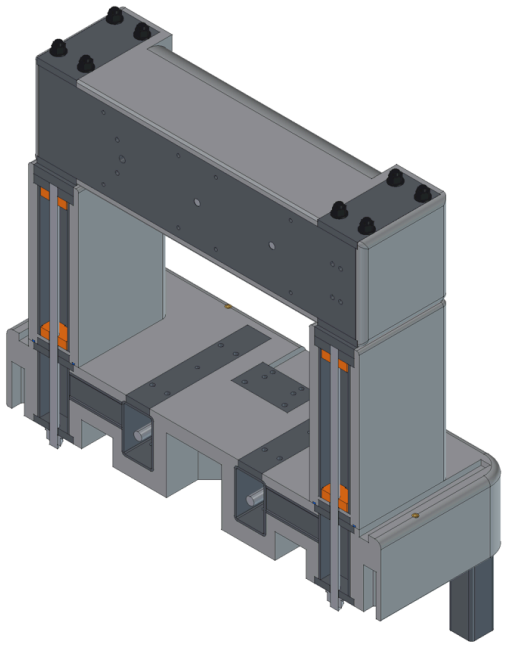

Top plate
Column (2x2 tube)
Foot bar



1. **Buy the steel cut to length.** `docs/1-sourcing/steel/README.md` has the cut list by stock size and every part (about 54 pieces, 46 kg). It's all common sizes: 3/8 in flat bar in 1-1/2, 2, 2-1/2 and 5 in widths, 1/2 x 2-1/2 in flat bar, 2 x 3 in rectangular tube, and 2 x 2 and 1-1/2 x 1-1/2 in square tube (1/8 wall). A local steel yard or Metal Supermarkets will cut it for you (cheapest); online sellers sell it in fixed lengths you cut with a chop saw or hacksaw. Plus, bought at length: 2 x 2 in tube, 4 x 8-1/8 in (the upright columns), and 1-1/2 x 1-1/2 in tube, 4 x 6 in (legs). A saw cut within 1/8 in is fine everywhere (the report's "Steel cut to length" table lists where each error goes).
2. **Drill and tap from the templates.** Print `docs/1-sourcing/steel/steel-templates.pdf` at actual size (check the 100 mm square), tape each part's outline to the part, center-punch every crosshair and drill on a drill press. The legend on each page gives the drill, and which holes to tap: 1/2-13 in the upright top plates' lower bars (the column tie rods screw in) and 3/8-16 for the beam studs; M5 in the base rail bars and the beam face bar; M6 in the support pads and the beam face bar. Mark one of each pair and use it as a drilling jig for the other.
3. **Bolt each skeleton together on the bench:**
   - **Upright** (two): just the parts for now: foot bar, two columns (a printed ring in each end) and the top plate's two bars. The column tie rods clamp it at assembly.
   - **Beam:** screw the three 107 mm studs into the face bar, a pipe nipple over each, the back bar over them and a nut on each; the seat and cap bars go in the mold on the cap-screw rods.
   - **Base** (one grid per half): each girder (2 x 3 tube) between its end plates, its rail bar bolted on top with 1/4-20 x 4 in bolts (nuts under the tube) and the draw rod through it; the pedestals (seat bar, the 2 x 2 tube spacers, bottom bar) held together with the tie rods for now; the cross tubes between the girders and from each girder to its pedestal, each clamped by its 1/2-13 rod and two nuts; the support pads bolted onto their cross tubes. Glue the four 3/8 x 1 in dowel pins into the rear half's split end plates, half their length out.
4. **Rail risers** (cold-rolled 1018, 1-1/4 in wide, ordered cut to the rail lengths): Y 3/4 in x 600 mm (2), X 5/8 in x 400 mm (2), Z 5/8 in x 300 mm (2). The rails are used at full length.
   - Clamp each rail on its riser, centered, and use the rail's own holes as the drill template; drill and tap M4 (60 mm pitch). Mark which rail goes on which riser.

- Y and X risers also get the hold-down holes, midway between rail holes and under the rail: 5.5 mm through with a 9.5 mm x 4.5 mm deep counterbore from the top. Positions from the riser center: Y at 0, ±120 and ±240 mm, X at ±30 and ±150 mm. They go into the tapped rail bars (base) and face bar (beam).
- Stone or lightly sand off burrs; keep the top faces flat.

5. **Clean all the steel:** wire-brush or flap-disc to dull gray metal, then degrease with acetone just before casting. If it will sit more than a day or two, give it a thin coat of epoxy primer (not on the faces the mold screws or the rails bear on). Paint only the part of each leg that will hang below the base (after casting).
6. **Threaded rod and hardware:** cut the 1/2-13 rods: per upright 2 x 363 mm (in the columns) and 1 x 153 mm (at the toe); 2 x 687 mm for the base draw rods; 4 x 270 and 6 x 225 mm for the base cross rods. Cut the 3/8-16 grade B7 rod into 8 beam studs of 186 mm and 3 ties of 107 mm. Deburr the ends. A rod a few mm off is fine: one end is set flush and the error ends up in the 6 mm tail past the far nut. Have 11 x M5 cast-in inserts ready for the enclosure (the only inserts left).



The cast-in steel for each base half (castings drawn see-through):

